

Renewable energy status and uptake in Kenya

Ibrahim Kipngeno Rotich^{a,b,*}, Hilda Chepkirui^c, Peter K. Musyimi^{d,e}

^a ELTE Eötvös Loránd University, Faculty of Science, Doctoral School of Environmental Science, Budapest, Hungary

^b ELTE Eötvös Loránd University, Savaria Institute of Technology, Budapest, Hungary

^c Jomo Kenyatta University of Agriculture and Technology, Department of Electrical and Electronics Engineering, Nairobi, Kenya

^d Karatina University, School of Education and Social Sciences, Department of Humanities and Languages, P.O BOX 1957-10101, Karatina, Kenya

^e ELTE Eötvös Loránd University, Institute of Geography and Earth Sciences, Department of Geophysics and Space Science, Budapest, Hungary

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ABSTRACT

With the increasing threat from global warming, renewable energy has been considered an alternative in several forums. Kenya is an emerging economy endowed with renewable energy resources due to its geographical location. This review aims to identify policies guiding the extraction, utilization, challenges, and projection of renewable energy. A systematic approach was used in the study by considering relevant journal articles and other gray documents such as Energy Acts and reports from international and national organizations on renewable energy. The findings show that the energy matrix in Kenya comprises 80 % fossil fuels, 18 % renewable energy and 2 % coal. Over 85 % of the total population utilizes wood biomass, with 86 % in rural areas and 21 % in urban areas. Over 70 % of the Kenyan landmass is Arid and Semi-Arid, which have potential renewable energy. Electricity production in Kenya is produced by approximately 90 % renewable energy but has a target of 100 % transition by 2020. KenGen and IPPs generate 62.97 % and 35.95 % of the electricity generated, respectively. Geothermal energy in Kenya is the highest source of electricity at approximately 45 %. The current amount of electricity connected to the grid is 2708 MW in 2021, with projections of 4763 MW, 6638 MW and 9790 MW in the low, reference and vision scenarios, respectively. By 2022, the percentage of Kenyan who had access to electricity was 76.89 %. It is estimated that, by 2100, the population in Kenya will reach between 80 and 220 million according to projection scenarios. An increase in populations leads to a greater energy demand, which is implicated in climate change. Based on the RCPs, the energy demand outlook shows that RCP2.6 will increase by 2–3 times, while RCP8.5 will increase by 6–8 times the current energy demand. The challenges experienced by renewable energy include inadequate financing, implementation hindrances, inadequate infrastructure and technological challenges, outdated policies, overambitious projects, landscapes affecting transmission, corruption, high operational costs, and energy stacking affecting the process. Regular reviews on energy policies and innovations provide data that can be used to strategize future energy equity and access while maintaining environmental sustainability.

1. Introduction

The African population has been steadily increasing, accounting for approximately 0.98 billion people globally. Population growth in Africa is projected to double in the next 50 years and triple in the next 100 years [1]. To meet basic needs in the coming years, such as healthcare, food security, shelter, and clothing, sustainable energy is a crucial factor. Although the African continent possesses substantial clean energy resources, the supply remains low, with one-third of the population (approximately 600 million people) lacking access to the grid [2,3]. Electricity penetration in sub-Saharan Africa (SSA) is less than 50 %,

compared to the world average of 90 % [4,5]. This distribution indicates that 32 % of the rural population and 78 % of the urban population have access to electricity [6]. Over the past decade, Africa's energy consumption has increased by 45 %, which has fostered economic growth but still falls short of meeting the population's supply and demand needs [7]. The United Nations launched Sustainable Energy for All (SEforALL) to promote universal electricity access, increase the global percentage of renewable energy from 18 % to 36 %, and enhance energy efficiency.

The country's energy mix is dominated by biomass at 62.5 %, petroleum products at 18.5 %, renewable energy resources at 16.9 %, and coal at 2.1 %. Approximately 80 % of the Kenyan population relies on

* Corresponding author. ELTE Eötvös Loránd University, Faculty of Science, Doctoral School of Environmental Science, Budapest, Hungary.

E-mail address: Ibrahimrkipng@gmail.com (I.K. Rotich).

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biomass, consuming up to 30 million tons per year [8]. It is estimated that the biomass supply and demand in Kenya was 15 million tons compared to 35 million tons as of 2002 [9]. With the growing population, the imbalance between demand and supply is expected to increase, with a projection of 54.01 million tons by 2020 [9]. This imbalance will strain forest cover, agricultural activities, and human settlements. To address this challenge, Kenya set a goal to improve household energy efficiency by increasing the use of cookstoves by 30 % [10]. However, this goal was not achieved due to several factors, including economic and financial challenges, lack of awareness, supply chain issues, and policy implementation challenges [11–13].

Petroleum import products increased from 12 % to 14 % between 2015 and 2017 [14], but this has an impact on macroeconomics, trade liberalization and energy sector which are prone to geopolitics which impacts the growth [15]. The dependence of petroleum products such motor and industrial gasoline, liquefied petroleum gases (LPG) and kerosene for transportation, industrial operation, domestic use, and power generation. However, this challenge has an impact on the environment due to the emissions which [16] concluded that it still understudied.

Electricity generation in Kenya has been predominantly driven by renewable energy, contributing 80 % of the total supply, with an estimated annual growth of 3.1 % [17]. Kenya has relatively low per capita electricity consumption, estimated at approximately at 190 kWh per year, compared to global average of roughly 3,200 kWh [18]. The cost of electricity persists at elevated levels approximately at \$0.20 per kWh which is higher compared to global average of about \$0.13 per kWh [19]. There has been plans to increase the electricity access and reduction of costs through initiatives such as Vision 2030, the BIG-4 Agenda, and universal electricity access through the Last Mile Connectivity Program aiming to ensure reliable and climate-resilient energy systems. The goal of achieving a 100 % transition to renewable energy by 2020 has laid a strong foundation for greener and more sustainable energy [20]. With electricity demand expected to triple in the next decade in East Africa, diversification is necessary to supplement and increase energy equity [21].

Kiplagat et al. (2011) [22] reviewed renewable energy potential in Kenya in 2010 but did not consider the impact of policies on future energy. Since then, there have been policies and energy acts, projections and bills on renewable energy that have been passed by Parliament which was aimed to be addressed in this review. Oluoch et al. (2021) [23] performed a perception study on renewable energy using a discrete method and analyzed the results using a conditional logit (MNL) and random parameter logit (RPL) from 1020 respondents. The considered input included the economic, social, and environmental impacts of renewable energy; however, based on the findings, they agreed on socioeconomic implications. In the environment, they agreed that renewable energy had a significant impact but did not specify how it would. Based on the policies provided, energy education seems to lag, and thus, there are challenges facing its implementation. Yusuf et al. (2024) [24] reviewed Liberia's energy landscape, policies, challenges, and opportunities since 2010 to enhance energy access. The findings show some similarities in terms of biomass used as a primary source between Liberia and Kenya. This situation creates a gap in the need for alternative energy sources.

Several gaps exist in the effectiveness of current renewable energy policies despite Kenya's significant effort. The gaps range from inconsistency and inadequate coordination between relevant regulatory frameworks and agencies. While there are policies in the Energy Act, 2019 promoting renewable energy, such as the Feed-in Tariff (FiT) policy, their implementation often suffers improper governance [25]. Another critical gap revolves around insufficient financial incentives and support mechanisms for renewable energy projects. Although licenses and current policies like FiT aim to provide guaranteed prices for renewable energy producers, the rates are not competitive to attract investment from private sector [26]. Lack of comprehensive fiscal

incentives, such as tax subsidies, lowers the initial capital expenditure for renewable energy projects. This gap is worsened by limited access to affordable monetary resources, especially for small and medium-sized enterprises (SMEs) involved in renewable energy. Furthermore, public awareness and capacity-building initiatives are critical to ensure stakeholders understand these policies effectively. Addressing these gaps is fundamental for enhancing the impact of renewable energy policies in Kenya and accelerating the Kenya's transition to a sustainable energy in future. This review article aims to present a comprehensive overview of Kenya's energy situation, detailing the different energy sources, the governing policies and regulations, the challenges to energy access, and the potential for renewable energy. By exploring these aspects thoroughly, the goal is to identify opportunities to enhance energy access and support sustainable development in Kenya.

2. Methodology

2.1. Methods used in renewable energy in Kenya

This review paper aims to assess the current knowledge on renewable energy and the goal of achieving 100 % adoption in Kenya by 2030. It presents up-to-date data and information on renewable energy penetration and potential, as well as future perspectives on policies and technical aspects in Kenya. This is accomplished through a systematic review of the relevant literature, guided by the question **"Renewable energy in Kenya"**. The inclusion and exclusion criteria involved the theme **"State of renewable energy in Kenya"** from selected articles from previous journals and gray literature sources such as the World Bank, Energy Acts and Policies, the Least Cost Power Development Plan (LCPDP), the Ministry of Energy, and sectoral frameworks. From the preliminary results over 683 peer reviewed journal articles were found which was later reduced to 167 from themed policies. However, after inclusion and exclusion criteria, 87 relevant journal articles with other gray materials from other sources published between 2001 and 2023. The tools used in finding the research articles were Google Scholar, Web of Science, Elsevier, Parliamentary Acts and Bills, other secondary document sources such as World bank, African Development Bank, IRENA, RenCon Associates limited among others.

2.2. Geographic pertinence

Kenya's location in East Africa provides a geographical diversity throughout its land mass of approximately 580,367 km². The country's landscape comprises of plains and plateaus, mountain ranges, and the Great Rift Valley, hence its unique topography. Recurring tectonic movements within the Great Rift Valley are a critical source of geothermal energy. Kenya is Africa's leading geothermal energy production country, with major geothermal plants in Olkaria and Menengai. The geothermal potential is estimated to exceed 10,000 MW [9,22]. In addition, Kenya benefits from high solar insolation solar energy from 89 % of its land mass which is arid and semi-arid shown in Fig. 1b [27]. This is particularly true in the northern and northeastern regions, due to their intense solar radiation and large-scale solar farms. Moreover, areas near Lake Turkana and the Ngong Hills experience strong and consistent winds, making them suitable for wind power generation. Lake Turkana Wind Power Project, the largest in Africa, significantly boosts Kenya's renewable energy capacity. The diversity of rivers provides hydroelectric power stations for instance on river Tana and other rivers which provide a substantial portion of Kenya's electricity. The potential for both small and large-scale hydroelectric projects remain high, particularly in the highland regions endowed with rivers.

2.3. Demographic and energy poverty characterization

Like other African countries, Kenya continues to face significant energy challenges, including issues of energy security, environmental

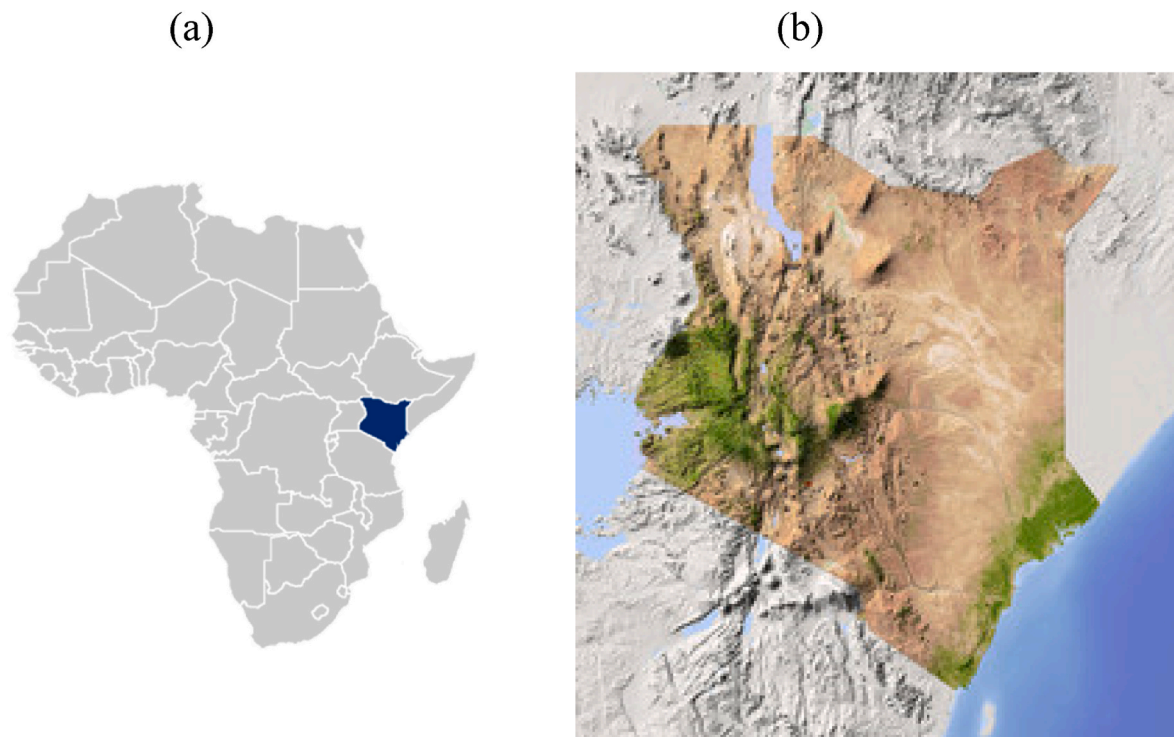


Fig. 1. The geographical position (a) of Kenya in Africa and (b) its elevation.

sustainability, and affordability. Energy equity and security play important roles at the population level based on social and economic pillars. Energy poverty in Kenya remains a significant challenge, characterized by limited access to modern energy services and reliance on traditional biomass for cooking and heating. According to the World Bank, as of 2020, approximately 75 % of the rural population in Kenya lacks access to electricity. In urban areas, this figure drops to around 23 %, highlighting a stark contrast between urban and rural energy access.

[28]. In urban areas, the classification can be further performed for high-density areas with perceived better infrastructure for the middle class and the high social class. However, slums experience severe energy poverty due to low income. Approximately 85 % of the population relies on biomass, such as wood, charcoal, and dung, for cooking [29]. This heavy reliance on biomass not only poses health risks due to indoor air pollution but also contributes to deforestation and environmental degradation. The World Health Organization (WHO) estimates that exposure to indoor air pollution from traditional cooking methods leads to approximately 21,560 deaths annually in Kenya [30].

3. Renewable energy status in Kenya

3.1. Biomass

This study extends the existing research on biomass energy derived from organic matter, which is converted to make electricity, fuel, and heat. The demand for biomass in Kenya is greater than the supply, posing a challenge to the already dwindling forest cover [22]. The government has promoted agroforestry to increase the stock of woody biomass to compensate for forest loss through a multidisciplinary approach by the Ministry of Agriculture, Energy, Environment, and Natural Resources.

3.1.1. Biofuels

Biofuels are fuel produced from organic matter such as plants, algae, or agricultural wastes over a short period of time [31]. The main biofuels include biodiesel, biobutanol, hydrotreated vegetable oils and fats and lignocellulosic-based fuels [32]. Substrates are converted into useable

fuel depending on the nature and methodology used, such as membrane technologies for bioreactors. In Kenya, the use of biofuel has not fully gained traction, with plans being put in place to construct bioethanol plants in 2020. Currently, several studies have been carried out to determine the potential of *Jatropha Curcas* [9,33,34], coconut, sorghum [35,36], croton [37], cotton seeds, and the diversification of ethanol from sugar companies [38,39]. A study by Ref. [40] in Western Kenya on the influence of biofuel production revealed that biofuel production was directly linked to food security, but the growth lacked specific biofuel policy frameworks. Inadequate production technologies, expertise and limited knowledge among stakeholders inhibit growth and deployment in the region [22]. showed that Kenya has high potential for the production of biofuels from sugar components, while [41] discussed how the government had established 500,000 acres of land for growing *Jatropha*, which would reduce the import of fossil fuel by 5 %.

3.1.2. Biogas

Biogas production involves the anaerobic process of producing flammable gases from biomass waste. It is estimated that the measured technological capacity of household biogas has a potential of 1.25 million. Biogas digesters are classified according to their operation ability, which includes single-stage continuous digesters, two-stage continuous digesters, batch digesters, fixed domes, floating drums and plug flow digesters. The biogas potential in Kenya is approximated to be between 20 and 131 MW [42]. Large-scale biogas plants are produced from slaughterhouses, agro-processing, and municipal wastes (landfills). To ensure food safety and waste management, several studies have been conducted on the extraction of biogas from invasive weeds. For example [43], used *Dodder Cuscuta* [44], used *E. crassipes* with slaughterhouse substrates, and [45] used dodder weed. *Opuntia stricta* was used in Laikipia County [46–48].

An innovative biogas-related company, for instance, the Flexi biogas company [49], aimed to expand biogas in Kenya using plug flow digesters. Several studies have investigated the utilization of biogas to improve sanitation, the use of biogas in transport (gas used in some tuktuks (rickshaw is used as a means of transportation), the conversion of biogas to be used in generators, domestic water heating and

agriculture (bioslurry). A study was performed using a two-stage continuous type from a fixed dome-shaped digester to plug flow, and more gas was extracted from the digestate (slurry of fixed dome).

[50] studied how energy poverty could be alleviated in rural households in western Kenya using biogas for cooking. It was established that cow dung is a highly available feedstock that contributes greatly to biogas production. The study recommended the use of inflatable digesters since they were economically and technically friendly. Most of the towns and cities in Kenya have designated slaughterhouses; however, most of the waste generated is not trapped. However, there are few slaughterhouses, such as [51,52], that directly use waste in biogas production. A study performed by Refs. [53,54] showed the potential of using animal waste from a slaughterhouse as a substrate and inoculum for biogas production.

3.1.3. Biomass cogeneration

Biomass cogeneration involves the heating of a fuel that is used in the simultaneous production of heat and power (CHP). These systems are found to be efficient in terms of power generation, waste management, and heat use in industrial setups, which can be converted to district heating and agricultural drying. Biomass cogeneration in Kenya has progressed through the use of organic waste or invasive weeds to produce electricity. The fuel sources for CHP plants include invasive weeds (Mathenge) [55], forest residues, organic waste, forest residues, and bagasse from sugar processing factories [56]. Based on sugarcane production, the energy from cogeneration in Kenya could produce 16.2 % of the total power capacity in Kenya [57]. [58] performed a techno-economic assessment of biomass cogeneration in KTDA factories using wood fuels. According to a previous study, the use of biomass generated from woodchips in the Finlays Saosa tea factory has shown potential since it began operation in 2009. Based on this study, biomass cogeneration from 58 existing factories could generate 30 MW of electrical power and over 264 MW of thermal power [39]. [59] studied the potential of biomass cogeneration by the Nzoia Sugar Company and established that the factory internally uses an effective capacity of 2.8 MW to crush the sugarcane generated from 7 MW installed. However, this can be increased to 15 MW through the modernization of boilers and turbine drives to increase efficiency. In Uganda and the western part of Kenya, bananas are farmed extensively. According to bioenergy studies conducted by Refs. [59,60], there is significant potential in using matooke peels, specifically Mbawazirume peel (MP) and Nakyinyika peel (NP), through pyrolysis and combustion processes. These findings indicate that the structure of peels, including cellulose, hemicellulose, and lignin, plays an important role in combustion. This process can potentially be replicated with other crops, offering a renewable energy source.

3.2. Geothermal energy

Geothermal energy is a resource obtained from radioactive elements on the crust that heats percolated water and converts steam, which can be used in the production of electricity [61–63]. Geothermal energy extraction depends on the type of steam present in a power plant, such as flash steam, dry steam, binary, hybrid, or enhanced geothermal systems. The energy potential in Kenya from geothermal energy is estimated to be 10,000 MW, but the exploited potential currently runs at approximately 8 % [9,64]. The Rift Valley boasts better production, with 14 sites mapped to have potential geothermal reserves. Currently, geothermal energy in Kenya has the highest share in the energy matrix, with approximately 45 % of total electricity production due to its highest capacity factor of over 95 % and its reliability [64,65]. Geothermal energy potential exploration in the region has been on a constant rise, with Kenya having the ambition of exploiting 5000 MW by 2030 and the medium goal of producing 1887 MW by 2017 [66,67]. A geothermal development company (GDC) is mandated to explore, develop, and produce geothermal resources through drilling in Kenya. Kenya

Electricity Company has a larger stake in the production and selling of electricity generated from geothermal energy, with the main sites being Olkaria in Naivasha.

3.3. Wind energy

Wind energy harnesses power by converting kinetic energy into mechanical energy, which is then converted to electrical energy. Wind energy is a source that has gained traction in the renewable energy sector in Kenya, but its potential is still high, with recent governmental policies such as a zero rating of wind energy components on developmental costs decreasing [68]. The short-term plan for wind power projects involves various stages of implementation, with some proposals pending the approval of FiTs. The best wind sites from the Solar and Wind Energy Resource Assessment (SWERA) show that the best sites are found in the ASAL regions with the Marsabit, Samburu, Meru, Laikipia, Meru, Nyeri, Nyandarua, and Narok Plains, which offer the best option for extraction, comprising approximately 90,000 km² of sites with wind speeds above 6 m/s [69]. Considering the intermittent correlation to underlying fluctuations in the wind speeds, the reliability of wind energy in the grid with capacity factors typically ranges between 20 % and 55 %. Studies using SWERA in 2008 indicate that the potential wind energy that can be harnessed in Kenya is approximately 4600 MW [92]. [70] performed a spatial wind site suitability study across the East Africa region using a geographic information system (GIS) and multicriteria and analytic hierarchy techniques. Kena was the best site, with 15.26 % and 1.55 % of the land mass found unsuitable for extraction, respectively. A study performed by Ref. [71] on the potential of wind energy in Taita Taveta and Garissa showed the potential of renewable energy potential with the IEC classification of class 2. However, this can be implemented to ensure that the KoSAP program is achieved by serving as alternative power generation for the surrounding communities. According to Ref. [72], cold regions exhibit optimal wind exploration performance [73]. used Weibull and Rayleigh distribution models to determine the wind potential in the Kisii region at a height of 10 m. The wind speeds at a height of 10 m were found to be 2.9 m/s with a standard deviation of 1.5. The calculated wind power density was 29 W/m². The extrapolation of the findings to 30 m showed the potential for operation in IEC class II [74]. characterized wind data in the Kesses region to determine the potential for wind energy at a height of 2 m from five years of data (2009–2013). The extrapolation was later performed to 10 m and 100 m. At a height of 10 m, the wind speed was 3.858 m/s, while at a height of 100 m, the wind speed was 6.926 m/s. The findings showed the potential of class II in the IEC, which can use mini grids.

3.3.1. VAWT potential and other uses

There have been few studies on VAWT potential in Kenya. According to Ref. [75] on unsteady VAWT potential using computational fluid dynamics (CFD), empirical results showed that Garissa was unsuitable for grid connections, while Marsabit showed potential [76]. performed an experimental study at 5 m/s to 10 m/s, and the results showed that nonuniform turbulent wind impacted the aerodynamic performance at low speeds, proving the self-starting ability of the system in the presence of turbulence. Most agricultural zones in Kenya have adopted the use of a windmill in the pumping of water for irrigation and community supply purposes and in tending livestock [77], telecommunications in regions with unreliable or limited grid supplies [78], desalination [79], street-lights, and research and development through wind patterns and exploring new potentials to improve performance.

3.4. Waste to energy (WTe)

The conversion of waste to energy can be achieved through various methods, including incineration, anaerobic digestion, gasification, and pyrolysis [80,81]. However, many urban areas in Kenya lack proper waste management practices, leading to significant amounts of waste

ending up at dumpsites. This waste includes solid waste such as food scraps, plastic containers, yard waste, and organic and inorganic waste from homes, businesses, manufacturing, agriculture, and slaughterhouses, among others, all of which can serve as fuel for waste-to-energy (WtE) processes. The amount of waste generated in Kenyan cities such as Kisumu, Nairobi, Nakuru, and Mombasa is estimated to be 1124 million tons per year, while other urban regions produce approximately 5.26 million tons annually [9,82].

Currently, there are few WtE plants operating in Africa, with many companies relying on existing facilities, as observed in Ethiopia [83,84]. Studies conducted by Refs. [85,86] explored the pyrolysis of old motor vehicle tires to produce petroleum products, aiming to reduce waste and decrease petroleum product imports [87]. numerically modeled the potential of municipal solid waste from the Dandora dumpsite. After 50 kg of waste was sampled, proximate and ultimate analyses were conducted, and the findings were utilized in Aspen and other numerical software for analysis. The results indicated the feasibility of generating 60 kW from gasification, suggesting that utilizing organic municipal waste could be financially viable for grid connections in Kenya.

3.5. Solar energy

Kenya boasts abundant sunlight throughout the year, presenting significant potential for harnessing solar energy. Its geographical location results in high insolation, with average peak sunshine hours ranging between 5 and 7 and an average daily insolation of 4–6 kWh/m² [88, 89]. Solar energy can be extracted at an efficiency rate of approximately 10–17 %, which can then be converted into heat (thermal) or through solar photovoltaic systems to generate electricity. The global horizontal irradiation (GHI) in Kenya is approximately 2400 kWh/m²/year, indicating substantial potential [90]. A viability study conducted by Ref. [91] on hybrid (a wind-solar resource) systems in Machakos revealed promising results. The solar insolation was measured at 2130 kWh/m², with a monthly average of 178 kWh/m². The reported daily insolation of 5.8 kWh/m² falls within the national insolation range, demonstrating the feasibility of solar energy utilization in the region. Another study conducted a technical performance analysis on the installed power system at Strathmore University over a one-year period in 2019, yielding a total power production of 734 MWh. The levelized cost of electricity (LCoE) and payback period were calculated to be US\$ 0.143/kWh and 9.1 years, respectively. The capacity factor was determined to be 13.97 %, with an annual specific yield of 1225 kWh/kWp,

aligning closely with results observed in existing power plants globally [92].

3.5.1. Potential concentration solar power (CSP) study performed in Kenya

Concentrated solar power (CSP) collects solar energy using heliostat mirrors that concentrate direct sunlight into a receiver. A higher DNI makes the CSP ideal, with rays being angled to be tracked and maximizing the direct angle with the sun. In Kenya, the direct normal irradiance (DNI) ranges between 1454 and 6649 Wh/m², with the western region having the lowest, while the northeastern region has a higher potential for concentrated solar power (CSP) [93,94]. The classification of the DNI in Kenya was studied by Ref. [95] using SWERA within a range of 1–10 suitability. The mountainous regions and regions in the western region had lower DNI values, while the remaining regions had DNI values ranging from 6 to 10, which fall within the suitable range of moderate to high DNI values. The regions found to have potential for CSP in Kenya were narrowed to Marsabit and Turkana Counties, with DNI values ranging from 670 to 1662 MWh [95], as shown in Fig. 2.

3.6. Hydropower

The hydropower potential is estimated to be approximately 6000 MW (small (3000 MW) and large plants (3000 MW)) with 818 MW connected to the grid. Hydropower is characterized by small hydro plants (minipower and micro power plants) (<3 MW), medium hydro plants (3 MW–30 MW), and large-scale plants (>30 MW). The hydrological sites in Kenya have been categorized into five geographical regions: the Rift Valley basin (345 MW), the Ewaso Ngiro basin (146 MW), the Lake Victoria basin (295 MW), the Athi River basin (84 MW), and the Tana River basin (790 MW) [96]. In the long term, the current plans for extracting large hydro water include Karura (90 MW) and High Grand Falls (500 MW) in the Tana catchment, Nandi Forest (50 MW) and Magwagwa (115 MW) in the Lake Victoria Basin and Aror (80 MW) in the Rift Valley Basin [97]. Kenya has been using runoff and impoundment systems. The studies on the potential of pumped storage in Kenya due to climate change effects (decline in rainfall patterns) and flood control downstream. Initial studies on pumped storage potential in Kenya showed its ability in Lake Turkana West, Samburu, Kapenguria, Kipcherere, Lomut, Sondu, and Homabay South. The hybridization of hydropower in Kenya has some potential, with studies performed on Kamburu, Kiambere, and Turkwell hydro plants to integrate solar power into existing infrastructure [98]. The coupling of hydropower and solar

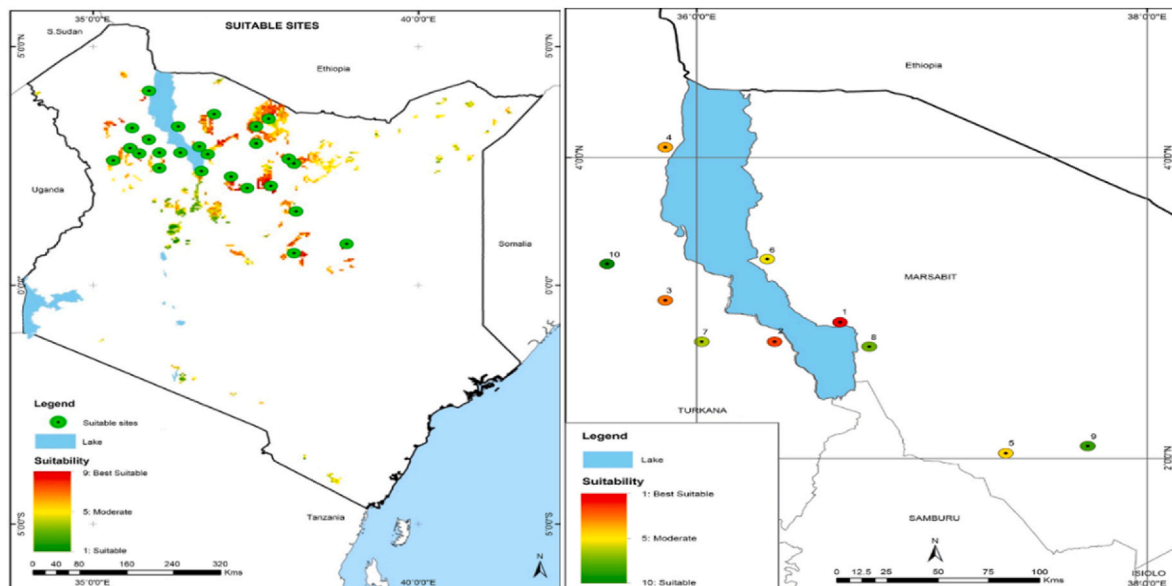


Fig. 2. Potential CSP power sites in Kenya [95].

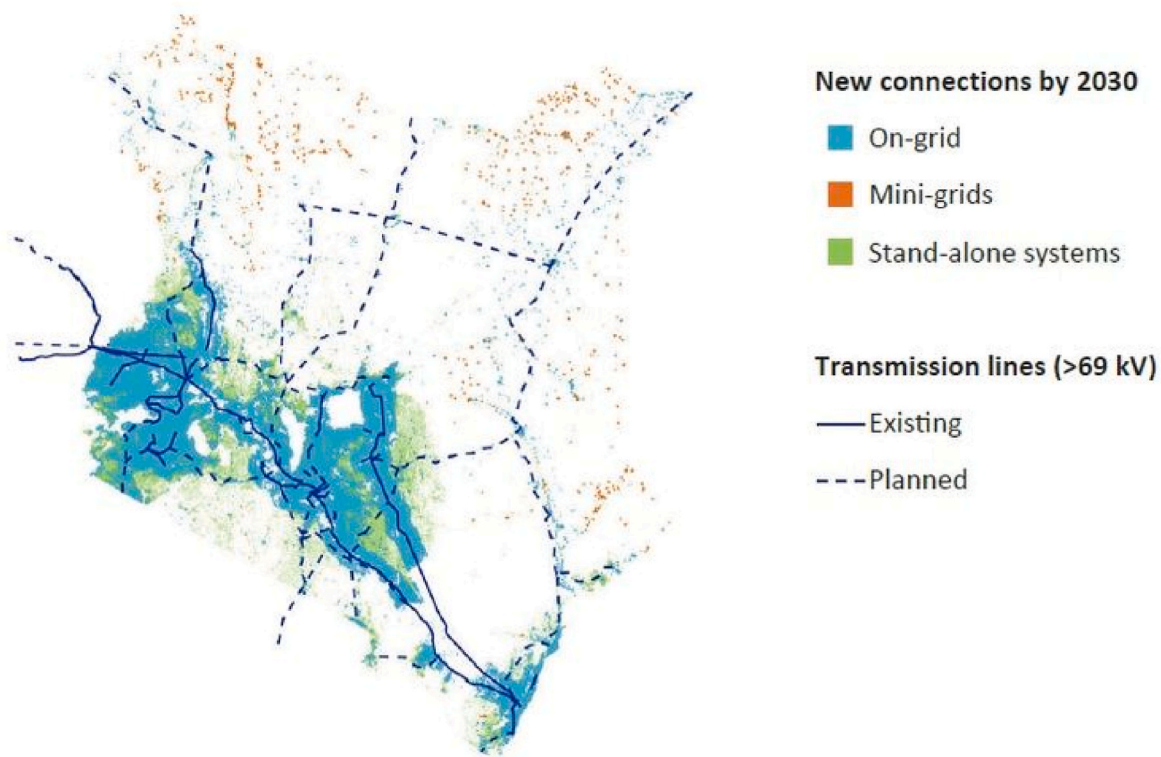


Fig. 3. Electricity expansion and distribution in Kenya. Source: <https://www.iea.org/articles/kenya-energy-outlook>.

systems helps to correct the negative correlation between solar and rainfall data (reduction in evaporation) and could help in water management.

3.7. Nuclear power

The Long-term Least Cost Power Development Plan (LCPDP) for the period 2017–2037 forecasts a peak energy demand ranging from 1937 to 9640 MW, driven by accelerated investment in the economy (see Fig. 3). The proposal to explore nuclear energy was first suggested in 2010 by the National Economic and Social Council (NESC) as a means to diversify the energy matrix. Kenya's plans for nuclear energy entail the installation of nuclear power plants by 2036, with considerations for plant safety, radioactive waste management, and proliferation concerns, particularly in light of terrorism challenges. The envisioned nuclear power infrastructure aims to incorporate four reactors, each with a capacity of 1000 MW, which is deemed necessary to meet electricity demands within the Eastern Africa Power Pool (EAPP). As of 2015, the global uranium reserves suitable for use as fuel were 5.9 million tons, with current consumption rates estimated to last for up to 130 years [99]. Although low levels of uranium oxide deposits have been discovered in Kenya, exploration efforts are ongoing to assess the feasibility of utilizing this resource.

4. Renewable energy strategies and implementation plan

The strategies that have been put in place to ensure the smooth transition include the expansion of the fuel base in the Parliament Acts; the implementation of existing policies and plans [100]; the enhancement of energy efficiency through the promotion of energy-efficient practices to optimize energy use, research, and development by understanding policy transfer with regions with similar policies [43,101]; the review of existing legal, regulatory, financial and circular economies [102]; and institutional frameworks to enhance sustainable generation, production, distribution and supply of electricity, as well as the engagement of communities in renewable energy projects. The

government needs to review the existing fiscal incentives for renewable energy projects and try to collaborate with other stakeholders to increase the economic and technical viability of renewable energy sources. Stakeholders need to be facilitated with fuel and look for a market for clean energy fuels, such as in e-vehicular systems, to increase demand.

4.1. Energy demand

The total interconnected electricity capacity in Kenya is projected to experience significant growth from the current 2708 MW in 2021–4763 MW in 2037 in the low scenario, 2103 MW–6638 MW in the reference scenario, and 2293 MW–9790 MW in the vision scenario. These forecasts are based on an upward trend observed in previous studies, with an annual growth rate of 3.92 % [69]. Over the past seven years, Kenya has experienced a remarkable annual growth rate of 19.14 %, attributed to accelerated electrification initiatives across the country. This growth is evidenced by the increase in the number of electricity customers, which increased from 3,611,904 customers in 2014 to 7,576,145 customers in 2020. This study expands upon existing research on GDP and commercial energy fluctuations, with a focus on their impact on economic performance, considering various energy scenarios. According to the 2030 goals in the medium scenario, it is anticipated that electricity and petroleum product demand will increase sixfold, with biomass demand reaching 53.41 million tons by 2020 [22]. However, demand has been a far-gone goal since the demand for biomass has increasingly led to the ban of logging, high electricity costs leading to several companies installing solar panels to generate their own power, frequent power outages leading to customers shifting to solar home systems, and costs of rural electrification being higher, leading to customers being unwilling to take power from the grid.

4.2. Energy production, transmission, and distribution

Electricity in Kenya is produced mainly by the Kenya Electricity Generating Company (KenGen) and Independent Power Producers (IPPS), which have 62.97 % and 35.95 % effective generation capacity,

respectively [103]. The effective electricity capacity in Kenya was 2708 in December 2020. On the other hand, the transmission in Kenya was performed by KETRACO (producer to the national grid), which was created under Sessional Paper No. 4 of 2004 to unbundle 5000 MW transmission, improve reliability, reduce transmission losses, and increase connectivity distribution functions. The total transmission length in 2019/2020 was 243,207 km, involving transmission (132 kV, 220 kV, and 400 kV) and distribution networks (66 kV, 33 kV, 11 kV, and 415/240 V). The reduction in system and reliability improvement led to a generation substation capacity increase from 3025 MVA in 2015–3878 MVA in 2020, and the distribution substations increased from 3572 MVA in 2015–4563 MVA in 2020, which significantly increased the distribution capacity from 6384 MVA in 2015–8174 MVA in 2020 [69]. Kenya Power is mandated in the bulk supply of electrical energy from the generation or transmission network through distribution lines and substations. The distribution lines comprise 46,051 km of 66 kV, 33 kV, and 11 kV lines with distribution substations and transformer capacities of 2800 MVA and 6195 MVA, respectively [69].

5. Challenges facing renewable energy in Kenya

The utilization of renewable energy has encountered several challenges. Renewable energy is significantly influenced by environmental factors, both directly and indirectly, depending on the situation and the specific resource, making it vulnerable to extremes [9,104]. Intermittent rainfall patterns in Kenya have resulted in a 9 % decline in hydropower generation [105,106]. Additionally, Kenya has experienced a loss of 7.6 % in forest cover, with demand outpacing supply, particularly affecting run-off hydro plants [107,108]. Farming in Kenya is the backbone of the economy, especially in the highlands. However, the demand for water for irrigation purposes introduces uncertainty regarding hydropower. With food security becoming increasingly essential for the population in Kenya, there is a growing need to produce more food, which in turn requires more water for irrigation. If this challenge persists, the dependability of hydropower may be at risk, as the available water supply could be affected in the future. To mitigate this risk, immediate measures need to be considered. These may include implementing hybrids of solar and hydro systems, establishing pumped water stations, and increasing forest cover to protect catchment areas. These actions are crucial for ensuring the consistent operation of hydropower and maintaining a reliable energy supply.

Renewable energy is affected by its availability and strength, causing intermittent and nonconstant energy production, which impacts the grid and necessitates energy storage solutions, especially during peak demand [109,110]. Solar, wind, and hydropower are seasonal, leading to lower power production in certain seasons. The instability of wind power on the grid presents challenges in the energy mix due to the high demand for reserve generation, which, if not managed properly, can lead to national blackouts. Solar energy is impacted by cloud cover, causing gaps and instability in the grid, while hydropower is susceptible to droughts, a situation worsened by the increasing threat of climate change. Given these developments in renewable energy, transitioning to 100 % renewable energy might be challenging, as deployment may require backup from other sources, such as fossil fuels, to fill the gaps in reserve capacity.

The lack of comprehensive policies, such as a regulatory framework and power purchase agreements, has limited feed-in tariffs, licensing, and financial and tax incentives [111]. There is also a lack of long-term planning, inadequate financial and institutional policies, and an absence of a legal framework to support the utilization, production, and distribution of liquid biofuels [112–114]. Biogas policy, which was intended to be developed in Kenya to reduce energy costs, indoor air pollution [30], and the burden on women and children, may be far from reality. This is evident from the failure of a similar policy in Rwanda, where most of the digesters failed [115].

In 2018, a project aimed to supply liquefied petroleum gas (LPG) to

vulnerable populations to cushion them from energy costs. However, this plan failed due to monopoly issues, safety concerns, and cost barriers, especially for low-income households. Payment plans affect feasibility, and delayed payments to suppliers contribute to setbacks [116–118]. Initially, solar and wind energy products were tax exempt, but a 14 % tax was reintroduced in Refs. [111,119], making these products expensive and limiting their penetration.

The evacuation of electricity from the Lake Turkana power plant to the national grid at Suswa faced issues due to weak financial policies, which led to the country signing a new contract that increased costs [120]. The project encountered challenges in securing financing and faced implementation delays. Electricity consumers in Kenya had to pay a \$52.5 million fine imposed on the government for the failure to evacuate electricity from the Lake Turkana Wind Power project [121]. While feed-in tariffs and net metering exist on paper, service providers face challenges due to weak policy implementation [122]. The National Construction Authority Act of 2012 failed in its implementation due to weak coordination between existing energy policies in Kenya [123–126].

The geographical location and terrain make power transmission challenging due to the need to acquire land and install infrastructure. The lead time from conception is extended due to the high resettlement costs associated with renewable energy pathways, such as wayleaves or communal lands. Theft and damage of energy products result in substantial costs for repairing or replacing stolen items, which can delay a project's financial viability [127]. Scattered population settlements in rural areas lead to long distribution lines, and the uncontrolled subdivision of arable land creates additional problems [128,129]. Harsh terrains, coupled with underdeveloped infrastructure, make certain areas inaccessible, leading to high costs for rural electrification projects. Distribution lines along coastal strips and major towns are prone to frequent interruptions due to corrosion. Poor physical planning fails to provide infrastructure corridors for electricity reticulation, resulting in high levels for purchasing land for renewable energy resource transmission.

Cultural effects, energy poverty, and societal concerns have contributed to a reduction in the transition to clean energy, resulting in the phenomenon of “energy stacking” and the “energy ladder.” These concepts stem from the intersection of access and equity issues arising from social and economic factors, leading to disparities in the adoption of renewable energy technologies [130–133]. The higher initial costs of renewable energy compared to traditional biomass limit access for low-income households. According to Ref. [30], rural homes rely heavily on kerosene lamps due to limited access to electricity. This reliance has adverse health effects, contributing to household air pollution (HAP), which in turn impacts the lives of school-going children.

Several pay-as-you-use companies, such as M-Kopa, have attempted to address this gap by introducing less expensive pico-solar and solar home systems (SHSs) to low-income communities. These systems serve limited purposes, such as lighting and powering radios. The recent government-led installation of grid power in primary schools has been instrumental in lighting underserved regions in rural areas. Areas near transformers were able to access electricity, significantly improving the lives of middle-class families and enhancing their social well-being. However, considering that most parts of Kenya are classified as arid and semiarid lands (ASALs), the communities living in these regions are often widely dispersed, making the installation of infrastructure costly and unattainable for many. While initiatives such as Lighting Africa have expanded electricity access to many regions, there remains a gap, particularly in connecting ASAL areas to the grid. This gap warrants further study and attention to ensure equitable access to electricity for all Kenyan communities.

5.1. Energy policies and acts in Kenya

Table 1 shows the energy policies and Acts in Kenya.

Table 1
Energy policies and acts in Kenya.

Energy Act/Policy		Content	Challenges	Implication	Gains
Electric Power Act	1964	Established to govern the generation, transmission, and distribution of electric power in Kenya.	Inadequate infrastructure Limited coverage High operational costs.	Led to the creation of the East African Power and Lighting Company (EAP&L).	Initiated the formal regulation and expansion of the electricity supply network.
Energy Act 2006	2006	Consolidated energy related laws Led to establishment of powers, and functions of the Energy Regulatory Commission (ERC) and Rural Electrification Authority (REA).	Implementation challenges Corruption Inadequate funding.	Increased regulatory oversight and the initiation of rural electrification programs.	Led to the opening of rural areas through electricity access. Established frameworks for renewable energy and energy efficiency.
Feed-in Tariff (FiT) Policy (2008, revised in 2012)	2012	Aimed at encouraging investors in renewable energy sources such as wind, solar, and biomass by guaranteeing fixed payments to renewable energy producers.	Lack/limited grid capacity to integrate renewable energy. Implementation challenges.	Stimulated interest and investment in renewable energy projects.	Led to installation of some FiTs projects such as Strathmore University which led to diversification of energy mix.
Kenya Vision 2030	2008	Outlined ambitious development goals for 2030, with energy being one of the supporting pillars through industrialization, transport, electricity distribution which would expand electricity generation to approximately 22000 MW and thus expanding energy access.	High project costs. Funding gaps Overambitious projects	Provided a long-term strategic framework that guided subsequent energy policies and investments.	Enhanced focus on renewable energy and energy efficiency initiatives.
Energy Regulatory Commission (ERC) Strategic Plan (2013–2018)	2013–2018	Focused on regulatory reforms, capacity building, and improving service delivery in the energy sector.	Implementation challenges	Enhanced regulatory practices and improved stakeholder engagement, although some goals were not fully realized.	Strengthened regulatory frameworks and improved compliance and enforcement. Enhanced institutional capacity and stakeholder engagement in the energy sector.
Energy Act (2019)	2019	Aimed at updating and consolidating energy sector laws, establishing three key regulatory bodies: the Energy and Petroleum Regulatory Authority (EPRA), the Rural Electrification and Renewable Energy Corporation (REREC), and the Nuclear Power and Energy Agency (NuPEA).	Coordination among the new regulatory bodies and addressing legacy issues from previous policies were significant challenges.	Improved regulatory framework and clearer mandates for energy governance, though operationalizing the new structures remains ongoing.	Streamlined and modernized regulatory frameworks. Clearer mandates and improved coordination among regulatory bodies. Enhanced focus on renewable energy, rural electrification, and potential future nuclear energy development.
Kenya National Electrification Strategy (KNES) (2018)	2018	Developed to provide a roadmap for achieving universal access to electricity by 2022, focusing on both grid and off-grid solutions.	Financial constraints, logistical issues in remote areas, and the high cost of infrastructure development were key hurdles.	Progress in expanding electricity access, particularly in rural areas.	Accelerated electrification efforts, particularly in rural areas. Significant increase in the national electricity access rate. Enhanced use of off-grid solutions, such as solar home systems, to reach remote areas.
Least Cost Power Development Plan (LCPDP)		This ongoing series of plans, updated periodically, outlines Kenya's strategy for power generation and infrastructure development to meet future energy demands cost-effectively.	Forecasting demand accurately and securing investment for planned projects are persistent challenges.	Provides a strategic planning framework.	Facilitate prioritization of least-cost energy projects, optimizing resource allocation.
National Energy Policy (2020)	2020	Addresses energy security, affordability, and sustainability. Emphasizes on diversification of energy sources, increased use of renewables, and enhanced energy efficiency.	Balancing the energy mix, ensuring reliable supply, and meeting environmental goals are ongoing challenges.	A comprehensive approach to energy sector development, aiming for a more resilient and sustainable energy system.	Promoted energy security through diversified energy sources. Enhanced focus on renewable energy and energy efficiency, aligning with global sustainability goals.
Geothermal Resources Act (1982)	1982	Regulates the exploration, development, and utilization of geothermal resources in Kenya. Provides a framework for licensing and management of geothermal projects. Establishes procedures for obtaining exploration and drilling permits.	High upfront costs for geothermal exploration and development. Technical challenges related to drilling and resource management. Environmental concerns and	Enhanced a regulatory framework for geothermal resource development.	Significant increase in geothermal power generation capacity.

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Table 1 (continued)

Energy Act/Policy		Content	Challenges	Implication	Gains
5000+ MW Policy (2013)	2013	Aims to increase Kenya's electricity generation capacity by over 5000 MW within 40 months. Focuses on developing a mix of energy sources including geothermal, wind, coal, and natural gas.	community resistance in geothermal-rich areas. Financial constraints and challenges in securing investment for large-scale projects. Infrastructure challenges such as grid capacity and transmission issues.	Increased pressure on regulatory bodies to expedite approvals and oversight. Emphasis on sustainable and diversified energy sources to meet ambitious targets.	Rapid expansion of Kenya's electricity generation capacity. Increased energy security and support for industrial growth. Diversification of the energy mix.
Africa Renewable Energy Initiative (AREI, 2015)	2015	Aimed at accelerating the adoption of renewable energy across Africa. Targets the installation of 10 GW of new and additional renewable energy generation capacity by 2020.	Mobilization of financial resources and investment for large-scale renewable projects. Technical and capacity limitations in implementing and managing renewable energy technologies.	Strengthening of regional cooperation and integration in the energy sector.	Acceleration of renewable energy adoption across Africa, contributing to climate change mitigation. Enhanced energy access and affordability, particularly in rural and underserved areas.
Energy Management Regulation (2012)	2012	Mandates energy audits for designated industrial, commercial, and institutional facilities. Promotes the adoption of energy efficiency measures and technologies. Sets requirements for energy management and reporting.	Adoption and implementation of energy efficiency measures by industries and institutions. Limited awareness and expertise in energy management practices. Financial barriers to investing in energy-efficient technologies.	Increased role for energy audits and performance tracking.	Enhanced energy efficiency across various sectors, leading to cost savings. Reduction in energy consumption and associated environmental impacts.

5.2. Energy scenarios by 2100

By 2100, Kenya's population is projected to range between 80 million and 220 million [134,135], depending on various projection scenarios: the low scenario (low fertility and high development), reference scenario (medium fertility and moderate development), and vision scenario (high fertility and low development). This population growth will significantly impact industrialization, climate change, and energy supply and demand, which are all crucial components for meeting developmental goals (SDGs). To ensure energy security and sustainability over the long term, proactive measures such as policy enhancements, infrastructure investment, and technological innovations are essential.

Considering the effects of climate change and its resilience on environmental sustainability, representative concentration pathways (RCPs) were examined, including RCP2.6, RCP4.5, RCP6.0, and RCP8.5, which represent different climate scenarios such as temperature increases, precipitation patterns, and extreme weather events. Under RCP2.6, energy demand is expected to increase by 2–3 times the current level, moderated by the extensive use of energy-efficient technologies and practices. RCP4.5 projects demand 3–4 times more demand, reflecting moderate efficiency gains and a diversified energy mix. RCP6.0 predicts demand growth of 4–6 times the current level, driven by high population growth, industrialization, and significant climate change adaptation needs. RCP8.5 anticipates demand growth of 6–8 times, driven by rapid population growth, industrialization, and extensive climate adaptation requirements [136,137].

Energy demand classification in Kenya also considers scenarios, and outlooks and recommendations factored based on projected demands. This study couples the factors affecting demand growth with its projection by 2100, providing guidance on the challenges and opportunities for energy planning and management in Kenya's future development. The consideration was done based on the energy demand provided by the projection studies (Tables 2 and 3).

6. Outlook

Table 2

Outlook from projection studies on energy scenarios and its implication by 2100 on different sectors.

	Low Scenario	Reference Scenario	Vision Scenario
Industrialization	Slow pace of industrialization due to limited investments and technological advancements.	Moderate industrial growth with gradual improvements in infrastructure and technology.	Rapid and sustainable industrial growth driven by advanced technologies and robust infrastructure.
Population growth	Continued rapid population growth leading to increased energy demand.	Population growth stabilizes due to socioeconomic developments, but energy demand continues to rise.	Managed population growth due to improved socioeconomic conditions, leading to more predictable energy demand.
Climate change	Limited mitigation efforts, resulting in significant impacts from climate change, including extreme weather events affecting energy infrastructure.	Incremental improvements in climate change mitigation, but significant challenges remain.	Strong mitigation and adaptation measures in place, significantly reducing climate impact on energy systems.
Energy supply and demand	Insufficient energy supply to meet rising demand, leading to frequent power shortages and reliance on traditional biomass.	Balanced energy supply and demand with a mix of traditional and renewable energy sources, but periodic	Ample and sustainable energy supply meeting demand through a highly diversified energy mix dominated by renewables.

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Table 2 (continued)

	Low Scenario	Reference Scenario	Vision Scenario
Developmental goals	Difficulty in achieving Sustainable Development Goals (SDGs), especially those related to energy access, poverty reduction, and economic growth.	shortages may occur. Progress toward SDGs, but some targets, particularly in energy access and sustainability, may not be fully achieved.	Achieving or surpassing SDG targets, particularly in energy access, sustainability, and economic growth.
Diversification	Limited diversification of energy sources, with heavy reliance on hydro and biomass, making the energy sector vulnerable to climate variability.	Improved diversification with increased investments in geothermal, wind, and solar energy, but fossil fuels still play a significant role.	Highly diversified energy portfolio with significant contributions from geothermal, wind, solar, and other renewables, minimal reliance on fossil fuels.

7. Recommendations

Table 3

Recommendations from projection studies on energy scenarios and its implication by 2100 on different sectors.

	Low Scenario	Reference Scenario	Vision Scenario
Policy Strengthening	Enhance policy frameworks to attract investments in the energy sector, particularly in renewable energy.	Scale up investments in renewable energy projects to reduce dependency on fossil fuels.	Develop and implement a comprehensive and forward-looking energy policy framework that supports innovation and sustainability.
Infrastructure Development	Invest in resilient energy infrastructure to withstand climate impacts and meet growing demand.	Expand and modernize the national grid to enhance reliability and accommodate renewable energy inputs.	Adopt integrated energy planning approaches that consider long-term demand projections, technological advancements, and environmental impacts.
Energy Efficiency	Implement nationwide energy efficiency programs to reduce demand pressures.	Integrate climate resilience strategies into energy planning and infrastructure development.	
Public–Private Partnerships	Foster public–private partnerships to drive innovation and investment in diverse energy projects.	Promote innovation in energy storage, smart grids, and other technologies to improve energy efficiency and reliability.	Establish sustainable financing mechanisms to support large-scale renewable energy projects and infrastructure development.
Capacity Building	Strengthen capacity building initiatives to improve local expertise in renewable energy technologies.	Strengthen regional energy cooperation to leverage shared resources and infrastructure.	Enhance community engagement and participation in energy projects to ensure inclusive and equitable energy access. Invest in education and training programs to build a skilled workforce capable of driving the energy transition.

8. Conclusion

Climate change is an emerging threat in the global arena, and renewable energy is being sought after to overcome these challenges. Renewable energy extraction involves various geographically related technologies. Kenya stands in a geographic position where renewable energy is extracted and utilized to reduce overdependence on fossil fuels. This review aims to identify the advantages of renewable energy and the challenges hindering its growth. The study solely relied on existing data from the literature and other reliable gray documents such as Parliament Acts and international bodies such as the World Bank and IEA. The data collected were mainly related to policies, government goals, targets and ambitions. The findings show that renewable energy is still underutilized, with the majority of the available resources having a less than 10 % extraction rate, notwithstanding goals such as ambitious Vision 2030, which would improve industrialization. Geothermal energy currently serves the grid, with approximately 40 % of the total energy mix supplementing hydropower affected by the dwindling forest

covers in Kenya. To increase renewable energy intake, some of the positive effects include tax incentives, the encouragement of public–private partnerships (PPPs), financial support to help investors, and the introduction of energy centers at the county level. Other challenges found to affect penetration include the vandalism of existing structures, which discourages investors; financial challenges; or delays in fund distribution, causing the project to be expensive and preventing the EROI from being evenly achievable and thus the LCoE from being expensive to break even. Policy implementation in Kenya was found to have not followed the latter, affecting the penetration and adoption of renewable energy and affecting investors' trust. The market for electricity in Kenya remains high due to the monopoly on distribution; thus, there is a lack of competition, leading to investors with factories needing high energy moving out due to the high cost. Renewable energy studies were evaluated, concentrating on milestones, goals, and accomplishments from renewable energy sources as well as ongoing challenges. These impediments include those related to financing, grid integration, policy implementation, and technology limitations, which limit the sector's potential. Renewable energy studies were evaluated, concentrating on milestones, goals, and accomplishments from renewable energy sources as well as ongoing challenges. Based on the findings, the applicability would help in assessing the effectiveness of specific policies and incentive programs which would promote renewable energy uptake. Exploring innovative financing mechanisms and strategies for grid modernization could address some of the identified challenges.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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